King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 208 Major Exam II Term 222

Original exam date: 27 March 2023. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. A linear homogeneous constant-coefficient equation which has the general solution $y(x) = (A + Bx)e^{2x} + Ce^{-2x}$ is

(a) $y''' - 2y'' - 4y' + 8y = 0$

(b) $y''' + 6y'' - 4y' + 8y = 0$

(c) $y''' - 2y'' + 4y' + 8y = 0$

(d) $y''' - 2y'' - 4y' - 8y = 0$

(e) $y'' - 4y = 0$

2. The rank of the matrix $A = \begin{bmatrix} 1 & -2 & -3 & -5 \\ 1 & 4 & 9 & 2 \\ 1 & 3 & 7 & 1 \\ 2 & 2 & 6 & -3 \end{bmatrix}$ is

(a) 1

(b) 2

(c) 4

(d) 0

(e) 3

3. An appropriate form of a particular solution y_p for the non-homogeneous differential equation

$$(D-1)^3(D^2-4)y = xe^x + e^{2x} + e^{-2x}$$

is given by $y_p(x) =$

(a) $Axe^{2x} + Be^{-2x} + Cx^3e^x + Dx^4e^x$

(b) $Axe^{2x} + Bxe^{-2x} + Ce^x + Dx^2e^x + Ex^3e^x$

(c) $Ae^{2x} + Be^{-2x} + Ce^x + Dxe^x$

(d) $Axe^{2x} + Bxe^{-2x} + Ce^x + Dxe^x + Ex^2e^x$

(e) $Axe^{2x} + Bxe^{-2x} + Cx^3e^x + Dx^4e^x$

4. Given that the vectors $v_1 = (2, 0, 0, 0)$, $v_2 = (0, 3, 0, 0)$, $v_3 = (0, 0, 7, 6)$ and $v_4 = (0, 0, 4, a)$ form a basis for $\mathbb{R}^4$, then $7a \neq$

(a) 12

(b) 18

(c) 24

(d) 0

(e) 6

5. Which one of the following sets V is a subspace of $\mathbb{R}^3$?

(a) V is the set of all (x, y, z) such that $y = 1$

(b) V is the set of all (x, y, z) such that $z \geq 0$

(c) V is the set of all (x, y, z) such that $z = 2x + 3y$

(d) V is the set of all (x, y, z) such that $xyz = 1$

(e) V is the set of all (x, y, z) such that $x + y + z = 3$

6. If $y = c_1 e^{ax} + c_2 e^{-x} \sin x + c_3 g(x)$ is the general solution of the differential equation

$$9y''' + 11y'' + 4y' - 14y = 0,$$

then $a =$

(a) $\dfrac{7}{9}$

(b) -1

(c) $\dfrac{1}{3}$

(d) 1

(e) $\dfrac{5}{9}$

7. Given that $y_p = u_1 \cos x - \cos x \sin x$ is a particular solution of the differential equation $y'' + y = \tan x$, then $u_1(x) =$

(a) $\sin x - \ln|\sec x + \tan x|$

(b) $\sin x + \ln|\sec x + \tan x|$

(c) $\cos x - \ln|\csc x + \cot x|$

(d) $\sin x - \ln|\sec x|$

(e) $\cos x + \ln|\cot x|$

8. Consider the vectors $\vec{t} = (4, 20, 23)$, $\vec{u} = (1, 3, 2)$, $\vec{v} = (2, 8, 7)$ and $\vec{w} = (1, 7, 9)$. If $\vec{t} = a\vec{u}+b\vec{v}+c\vec{w}$, then $a + b - c =$

(a) 1

(b) -1

(c) -2

(d) 0

(e) 2

9. The dimension of the subspace consisting of the set of all vectors of the form (a, b, c, d) for which $a + 2b = c + 3d = 0$ is

(a) 1

(b) 4

(c) 2

(d) 3

(e) 0

10. If $y = c_1 \cos(\ln x) + c_2 \sin(\ln x)$ is a solution of the initial-value problem

$$x^2y'' + xy' + y = 0, \quad y(1) = 7,\ y'(1) = 2,$$

then $c_1 + c_2 =$

(a) 5

(b) 9

(c) -9

(d) 0

(e) -5

11. Which of the following sets of solutions of homogeneous linear differential equations, is linearly independent?

(a) $\{17,\ \cos^2 x,\ \cos 2x\}$

(b) $\{e^x,\ \cosh x,\ \sinh x\}$

(c) $\{1,\ e^x,\ \sinh x\}$

(d) $\{17,\ 2\sin^2 x,\ 3\cos^2 x\}$

(e) $\{0,\ \sin x,\ e^x\}$

12. The vectors $v_1 = (1, 0, 1)$, $v_2 = (2, -3, 4)$ and $v_3 = (3, 5, a)$ are linearly dependent if $3a =$

(a) -1

(b) 2

(c) 1

(d) 0

(e) -2

13. The general solution of the differential equation $y^{(4)} + 3y'' - 4y = 0$ is $y =$

(a) $c_1 e^x + c_2 e^{-x} + c_3 \sin 2x + c_4 \cos 2x$

(b) $c_1 e^x + c_2 x e^x + c_3 \sin 2x + c_4 \cos 2x$

(c) $c_1 e^{-x} + c_2 x e^{-x} + c_3 \sin 2x + c_4 \cos 2x$

(d) $c_1 e^x + c_2 e^{-x} + c_3 \sin 2x$

(e) $c_1 e^x + c_2 e^{-x} + c_3 \sin 4x + c_4 \cos 4x$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	Roots $2, 2, -2$: $(r-2)^2(r+2) = r^3 - 2r^2 - 4r + 8$.
2	**(e)**	Row reduction leaves 3 nonzero rows (4th row duplicates the 2nd): rank 3.
3	**(e)**	Roots $1, 1, 1, 2, -2$: trial terms xe^{2x}, xe^{-2x}, x^3e^x, x^4e^x.
4	**(c)**	$\det = 6(7a - 24) \neq 0$, so $7a \neq 24$.
5	**(c)**	Only $z = 2x + 3y$ is a homogeneous linear condition; the others miss $\mathbf{0}$ or fail closure.
6	**(a)**	$9r^3 + 11r^2 + 4r - 14 = (r^2 + 2r + 2)(9r - 7)$, real root $a = \frac{7}{9}$.
7	**(a)**	$W = 1$, $u_1' = -\sin x \tan x = \cos x - \sec x$, so $u_1 = \sin x - \ln\lvert\sec x + \tan x\rvert$.
8	**(d)**	Solve $a + 2b + c = 4$, $3a + 8b + 7c = 20$, $2a + 7b + 9c = 23$: $(a, b, c) = (5, -2, 3)$, so $a + b - c = 0$.
9	**(c)**	Two equations on four variables, b, d free: dimension 2.
10	**(b)**	$y(1) = c_1 = 7$, $y'(1) = c_2 = 2$, so $c_1 + c_2 = 9$.
11	**(c)**	Others contain 0 or satisfy an identity; $1, e^x, \sinh x$ is independent.
12	**(a)**	$\det = -3a - 1 = 0$, so $3a = -1$.
13	**(a)**	$r^4 + 3r^2 - 4 = (r^2 + 4)(r^2 - 1)$: roots $\pm 1, \pm 2i$.